Hanxin Wang

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Summary

A Ph.D. student researching computer vision with extensive experience in **few-shot adaptation with foundation models and class incremental learning**. Familiar with **image classification, image segmentation, and activity recognition**. Driven by a deep passion for acquiring new scientific skills.

Education

University of Macau Aug. 2024 - Present

Ph.D student in Computer Science

Research Interests: computer vision and machine learning, specifically focusing on robust foundation model adaptation with limited data and auto-annotation with expert-crafted guidelines on domain-specific tasks.

University of Electronic Science and Technology of China Sep. 2021 - Jun. 2024

M.S. in Information and Communication Engineering

Research Interests: computer vision and machine learning, specifically the application of continual learning to multi-modal activity recognition.

Fuzhou University Sep. 2017 - Jun. 2021

B.E. in the Internet of Things (Major) and Artificial Intelligence (Minor)

Received postgraduate recommendation

Research Experience

Enabling Validation for Robust Few-Shot Recognition

University of Macau *Oct. 2024 - May. 2025*

Background:

- Despite their strong few-shot adaptation capabilities, pretrained Vision-Language Models (VLMs) often struggle when encountering out-of-distribution (OOD) test data that presents a distribution shift from the in-distribution (ID) training examples.
- The core challenge of Few-Shot Recognition (FSR) is data scarcity, extending beyond limited training data to a complete lack of validation data

Contribution:

- Established the robust FSR problem and comprehensively evaluated existing VLM-based FSR methods.
- Proposed a novel validation strategy by exploiting the few-shot ID training data and external data, allowing hyperparameter tuning and checkpoint selection for better generalization to both ID and OOD test data.
- Presented a robust finetuning pipeline with our validation strategy, significantly outperforming existing VLM-based FSR methods w.r.t both ID and OOD accuracy.

Regularized Fusion for Continual Egocentric Activity Recognition

University of Electronic Science and Technology of China *Mar. 2023 - Jan. 2024*

Background:

- Existing continual learning methods hardly acquire discriminative multimodal representations of activity classes from different isolated stages.

Contribution:

- Designed a dynamic expansion fusion architecture to ensure the data within each stage can learn the optimal multi-modal representation.
- Introduced a confusion mixup regularized multimodal fusion network that can capture the dynamic change of correlation from different modalities and help alleviate the confusion between the in-stage data and out-stage data in the feature space.

- Extended the confusion mixup to a generalized adversarial architecture. This extension explicitly generates confusion samples via a learnable approach, further alleviating the confusion between the in and out-stage data in the feature space.
- The proposed method outperforms both SOTA unimodal and multimodal methods on existing multimodal continual learning benchmarks for egocentric activity recognition.

Multi-modal Egocentric Activity Dataset for Continual Learning

University of Electronic Science and Technology of China *May. 2022 - Jan. 2023*

Background:

- Continual multi-modal egocentric activity recognition is highly desirable in practical applications, however, it has not been thoroughly explored due to the lack of relevant datasets.
- Recently, the rapid development of wearable devices has made it easier to gather extensive collections of egocentric data.

Contribution:

- Collected multi-modal egocentric data with self-developed glass which contains synchronized data of video, accelerometer, and gyroscope for 32 types of daily activities.
- Proposed a benchmark model for multi-modal egocentric activity recognition and demonstrated the performance by evaluating the three modalities independently and in combination.
- Assessed the issue of catastrophic forgetting in continual multi-modal egocentric activity recognition and employed well-known continual learning strategies to address this problem.

Publications

- **Hanxin Wang**[†], Tian Liu[†], Shu Kong[*]. Enabling Validation for Robust Few-Shot Recognition. Arxiv, 2025.
- Shuchang Zhou[†], **Hanxin Wang**[†], Qingbo Wu[*], Fanman Meng, Linfeng Xu, Wei Zhang, Hongliang Li. Adversarially Regularized Tri-Transformer Fusion for continual multimodal egocentric activity recognition. Displays, 2025.
- **Hanxin Wang**[†], Shuchang Zhou[†], Qingbo Wu[*], Hongliang Li, Fanman Meng, Linfeng Xu, Heqian Qiu. Confusion Mixup Regularized Multimodal Fusion Network for Continual Egocentric Activity Recognition. ICCV Workshops, 2023.
- Linfeng Xu[*], Qingbo Wu, Lili Pan, Fanman Meng, Hongliang Li, Chiyuan He[‡], **Hanxin Wang**[‡], Shaoxu Cheng[‡], and Yu Dai[‡]. Towards continual egocentric activity recognition: A multi-modal egocentric activity dataset for continual learning. IEEE Transactions on Multimedia, 2023. ([‡] Student Author)

Workshops

- AutoExpert Workshop: 1st@CVPR'26
- Personalized Visual Intelligence Workshop: 1st@ACCV'26

Honors and Awards

The 5th *Place* in the campus women's group of Sichuan University Light Volleyball Competition	May. 2023
The 3rd *Prize* academic excellence scholarship of UESTC	Nov. 2022
Outstanding Graduates of Fuzhou University	Jun. 2021
Excellent Student Cadre of Fuzhou University	May. 2020
Provincial 3rd *Prize* in National Undergraduate Electronic Design Competition	Sep. 2019
Fuzhou University Tiandixing Scholarship	May. 2019
Merit Student of Fuzhou University	May. 2018
The 1st *Prize* comprehensive scholarship of Fuzhou University (three times)	Sep. 2017 - Jun. 2020